

# Lily Lab: Teacher Notes and Answer Key

For the student lab (7 pages) and the accessible IPP version (5 pages).

## At a Glance

|                      |                                                                                                                                                                                                                                                                        |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Outcomes</b>      | <b>LO 7-2-04</b> (flower structure, pollination, fertilization, seed formation). Interleaved: <b>LO 7-2-03</b> (Q8, photosynthesis) and <b>LO 7-2-05</b> (★, selective breeding).                                                                                      |
| <b>Where it fits</b> | Teacher guide <b>Lesson 8</b> (flower structure). Run it after Notes Section 4 introduces the four parts, and before Exit Ticket 4.                                                                                                                                    |
| <b>Grouping</b>      | Groups of 3–4. One lily + one lily leaf per group. Roles are on page 1 of the student lab.                                                                                                                                                                             |
| <b>Time</b>          | <b>Day 1:</b> 60–70 min. <b>Days 2 and 4:</b> 5 min (Press Managers only). <b>Mount day:</b> 45–50 min, about 10–14 days later.                                                                                                                                        |
| <b>Evidence base</b> | Retrieval before instruction (From-memory table, revisited at the end) · worked model before practice (labelled lily + finished-tray model) · refutation of "flowers are for people" (predict → evidence → revisit) · interleaved prior content (Q8, ★) · self-rating. |

## Shopping and Prep

| Item                   | Amount (class of 30, 8 groups) | Notes                                                                                                                                                                                                                                                         |
|------------------------|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Lilies</b>          | 8 stems + 2 spare              | Asiatic or Oriental (e.g. Stargazer) from a grocery store or florist. Buy <b>3–5 days ahead</b> and keep them in water in a warm room so the blooms are fully open. <b>Ask the florist to leave the anthers on;</b> many shops pull them to prevent staining. |
| <b>Archival paper</b>  | 16 sheets + spares             | Acid-free, lignin-free paper for the press sandwich (2 per group).                                                                                                                                                                                            |
| <b>Archival card</b>   | 8 (or 30, see flag)            | Acid-free cardstock for mounting, letter size or larger.                                                                                                                                                                                                      |
| <b>Attaching</b>       | 1–2 bottles / rolls            | pH-neutral (archival) PVA glue, or gummed linen / archival tape. <b>No regular tape or glue stick.</b>                                                                                                                                                        |
| <b>Blotter</b>         | ~64 sheets                     | Blotter paper, newsprint, or plain (unprinted) paper towel. 4 per group for Day 1, plus 4 per group for each swap.                                                                                                                                            |
| <b>Other</b>           | per group                      | 2 pieces of cardboard, tray or paper plate, small scissors, hand lens, clear tape, paper towel, gloves available, heavy books.                                                                                                                                |
| <b>Teacher station</b> | 1                              | Scalpel or sharp paring knife + cutting board for ovary cross-sections. Cut slices <b>2–3 mm thick</b> , across the ovary.                                                                                                                                    |

### ■ YOUR DECISION: ONE CARD PER GROUP, OR ONE PER STUDENT?

The student lab is written for **one mounted card per group**, because one lily only gives 3 sepals and 3 petals. Lily stems usually carry 3–5 blooms. If you give each group a whole stem, every student can press and mount their own card, which makes the card a stronger individual assessment piece. You'd need about 30 archival cards, and roughly double the archival paper and blotter.

### ⚠ SAFETY

- **Cats:** every part of true lilies (*Lilium*) and daylilies (*Hemerocallis*) is toxic to cats, including pollen and vase water. A small amount can cause kidney failure. Don't send lily parts home with students who have cats, and consider a note home.
- **Pollen:** stains skin, clothing, and paper. Lift it with tape or a dry tissue; don't rub, and don't wet it first. Ask about pollen allergies before Day 1 and have gloves ready.
- **Blades:** students use small scissors only. The teacher cuts all ovary sections.
- No eating. Wash hands at the end of Day 1 and of mount day.

## Run of Show

| When         | What happens                                                                                                                                              | Student pages        |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| Day 1 • 0–8  | Predict (who is the lily for?) and the From-memory table. <b>No notes.</b> Circulate and note what students can't recall; don't correct yet.              | p1 predict, p2 table |
| 8–13         | Safety rules, roles, read "The Lily Trick" together, point out the model and the finished-tray model.                                                     | p1–3                 |
| 13–40        | Dissect outside-in and record. Run the ovary-slice teacher station on a rolling basis: groups bring the pistil when they reach step 8.                    | p3 steps, p4 record  |
| 40–55        | Build the presses. Check that no parts touch and anthers were tapped before the top archival sheet goes on.                                               | p5                   |
| 55–65        | Clean-up and hand washing. Q1–3 finished in class; Q4–5 can be an exit task.                                                                              | p4–5                 |
| Days 2 and 4 | Press Managers swap the <b>blotter only</b> (never open the archival sheets) and fill in the press log. In a humid room, swap daily for the first 3 days. | p5 log               |
| Mount day    | 0–5: check dryness. 5–30: plan, mount, label. 30–50: Q6–9 and ★, finish the From-memory column, self-rating.                                              | p6–7                 |

## Pressing a Lily: What to Expect

Lilies are **fleshy**, so they press less neatly than thin garden flowers. That is useful for Q4. Parts are ready when they feel papery and room temperature. Damp parts feel cool. Allow **10–14 days**, and up to 3 weeks for thick parts.

| Problem                                 | Cause and fix                                                                                                                                                                 |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Black or fuzzy spots (mould)</b>     | Blotter stayed wet. Swap more often, keep the stack in a dry spot away from sinks, and use spare parts to re-press.                                                           |
| <b>Tepals turn brown or see-through</b> | Normal for lilies: they hold a lot of water and dry slowly. Faster, more frequent blotter swaps keep more colour. Photograph the fresh tray on Day 1 so students can compare. |
| <b>Orange stains on archival paper</b>  | Loose pollen. Tap the anthers well, or press the stamens on their own small archival slip.                                                                                    |
| <b>Ovary slice shrivels</b>             | Expected. It is mostly water. Keep slices 2–3 mm. The sketch on p4 is the real record of the chambers.                                                                        |
| <b>Style stays rounded</b>              | Thick and stiff. It still mounts fine. Splitting it lengthwise helps it lie flat.                                                                                             |
| <b>Parts crack on mount day</b>         | Very dry parts are brittle. Slide a thin paper strip underneath to lift them. Don't use fingers or tweezers on the tepals.                                                    |

## Formative Signals: What to Listen For

| If you hear...                                            | Then...                                                                                                                                                                      |
|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| "The flower is pretty for us."                            | Expected at the predict step. Don't correct it; Q9 does the refuting with the students' own evidence. Prompt with "Who else visits flowers?"                                 |
| Pollination and fertilization used as the same thing (Q7) | The most common error. If more than a few students do this, reteach the boundary (pollen <i>lands</i> on the stigma vs. pollen <i>joins</i> the ovule) before Exit Ticket 4. |
| Stamen and pistil swapped                                 | Anchor it on the tray: the powdery orange part makes pollen, the one in the middle with the green swelling makes seeds.                                                      |
| "The seeds come from the stem / soil."                    | Send them back to the ovary slice: "Where did you see ovules?"                                                                                                               |

## Answer Key: Student Lab

### Page 1: Predict

Accept any prediction with a reason. It gets revisited in Q9. Target idea: the colour, spots, and scent are for **pollinators** (insects such as bees, butterflies, and moths).

### Page 2: From memory (matches Notes Section 4)

|               |                                                                                      |
|---------------|--------------------------------------------------------------------------------------|
| <b>Sepal</b>  | Protects the flower while it is still a bud.                                         |
| <b>Petal</b>  | Attracts pollinators (colour, pattern, scent).                                       |
| <b>Stamen</b> | Male part. The anther makes pollen; the filament holds it up.                        |
| <b>Pistil</b> | Female part. The stigma receives pollen; the ovary holds ovules, which become seeds. |

## Page 4: Record table

| Part    | # | Where                          | Likely observations                                                                                                       |
|---------|---|--------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Sepals  | 3 | Outer ring, attached lowest    | Coloured like the petals; slightly narrower; may have a greener ridge or tip on the outside; green nectar groove at base. |
| Petals  | 3 | Inner ring, between the sepals | Slightly wider; same colour and spots; nectar groove at base.                                                             |
| Stamens | 6 | Around the pistil              | Long thin filament; anther orange-brown, powdery or sticky, swings on the filament.                                       |
| Pistil  | 1 | Centre                         | 3-lobed stigma, sticky; long style; green swollen ovary at the base.                                                      |

**Observation boxes:** pollen is orange-brown, sticky-powdery dust. The stigma feels sticky or wet. The ovary slice shows **3 chambers**, with pale, bead-like ovules in rows (usually 2 per chamber in one slice).

### 1. How did you decide which tepals were sepals?

They are the **outer ring**: attached lower and further out, and they covered the outside of the bud. They are often a little narrower, sometimes with a greener tip or ridge.

**Full marks:** at least one observable piece of evidence tied to position. "They looked different" alone is not enough.

### 2. How the stigma felt, and why that helps.

Sticky. Pollen grains brushed off a pollinator stick to it and stay there, so pollen is caught and held where it is needed.

### 3. Estimate the seeds; what must happen first?

Accept any reasoned estimate built from the count. For example, ovules per slice × about how many slices fit along the ovary. A real lily ovary holds dozens to a few hundred ovules. First: **pollination** (lily pollen reaches the stigma), then **fertilization** (pollen joins with the ovule).

**Full marks:** shows how the estimate was made and names both steps.

### 4. Which part will press best and which worst?

Best: thin parts with little water, such as the leaf, a filament, or the thin edge of a tepal. Hardest: the ovary or pistil (thick and full of water). The fleshy base of the tepals is hard too.

**Look for:** reasoning that links thickness and water content to how easily a part dries flat.

### 5. Why swap the blotter?

The blotter soaks up water until it is full. A wet blotter can't pull out any more, and it keeps the parts damp, so they brown or go mouldy. Dry blotter keeps drawing water out.

### 6. Why acid-free paper?

Acid slowly breaks down paper and the specimen: it yellows, turns brittle, and can damage the plant. Acid-free paper keeps the specimen whole for a very long time, so future scientists can still study it. For example, they can compare plants across decades to see how species or ranges have changed.

**Full marks:** links the acid to damage over time, and the damage to losing a scientific record.

## 7. Trace a pollen grain.

**Anther** makes pollen → pollen sticks to the bee → bee visits another lily → pollen rubs onto the sticky **stigma**.

*Pollination ends here: pollen has landed on the stigma.* → pollen grows a tube down the **style** → reaches an **ovule** inside the **ovary** → joins with the ovule. *This is fertilization.* → the ovule becomes a seed and the ovary becomes the seed pod (fruit).

**Scope:** "pollen tube" and "sperm/egg" are not required at this level. Accept "pollen travels down the style to the ovule." The key is putting the boundary in the right place.

## 8. Which part makes the most glucose, and where did the flower's energy come from?

The **leaf** (green, with chloroplasts). Accept the green ovary as a minor second. The leaves used photosynthesis (light + water + carbon dioxide → glucose + oxygen) to make glucose. The phloem carried that glucose to the flower to build it. The coloured tepals do little or no photosynthesis.

**Interleaves LO 7-2-03.** Watch for "the flower got energy from the soil", which is the unit's core misconception resurfacing.

## 9. "Lilies are big and colourful so people will buy them."

**Disagree.** Evidence from the dissection: bright colour and spots that pollinators can see (spots lead toward the nectar); strong scent (many Oriental lilies are moth-pollinated and smell strongest at night); nectar grooves at the base of each tepal as a reward; anthers held out on long filaments where a visitor brushes them; sticky stigma in the visitor's path. Wild lilies had these traits long before people sold flowers.

**Full marks:** two pieces of dissection evidence, each linked to pollinators. Accept "partly agree" if the student separates the original purpose (pollinators) from what breeders later exaggerated.

## ★ Selective breeding

**People probably made bigger or brighter:** bloom size, new colours, more buds per stem, longer vase life, even pollen-free varieties. **There first for pollinators:** colour, spots, scent, nectar, and the position of the stamens and stigma.

**Interleaves LO 7-2-05** (selective breeding = people choosing which plants reproduce).

## Answer Key: IPP Version

|                      |                                                                                                                          |
|----------------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>p1 guess</b>      | Any answer. Revisited on p5.                                                                                             |
| <b>p3 counts</b>     | Sepals 3 · Petals 3 · Stamens 6 · Pistil 1                                                                               |
| <b>p3 circle-its</b> | Orange pollen: <b>stamen</b> . Top of pistil feels <b>sticky</b> . Why: "it catches pollen" (said, pointed, or written). |
| <b>p4 circle-its</b> | <b>To dry them flat</b> . Hardest: <b>thick pistil</b> . Why: "it is thick" or "it has lots of water".                   |
| <b>p5 match</b>      | Sepal → covers the bud · Petal → calls bees and butterflies · Stamen → makes pollen · Pistil → makes seeds               |
| <b>p5 revisit</b>    | <b>Bees and butterflies.</b> "How do you know?": bright colour, spots, smell, or nectar.                                 |

### ★ DISPLAY IDEA

Mounted cards make a strong class "herbarium" wall or binder. If you keep them, store them flat in a folder away from light, and the colours will last years. Label each card with the date, so next year's class can see how the specimens held up.